

AIE683 Machine Learning

Week 1: Introduction & Tooling

Dr. Hakan Emekci

Hacettepe University, Graduate School of Informatics

Logistics

- **Lectures:** Thursdays 08:40–11:40
- **Office hours:** Thursdays 12:00–13:00 and 16:00–17:00
- **Email:** hakanemekci@hacettepe.edu.tr
- **Materials:** <https://github.com/hemekci/AIE683-Machine-Learning>

Assessment

Component	Weight
Paper proposal	10%
Homework assignments	20%
Project & research paper	70%

The project is a research paper: proposal → progress report → presentation → final paper with code.

Academic Integrity

- Homeworks and projects are checked for plagiarism
- Copied work results in 0 points for everyone involved
- You may reuse code from the course slides and notebooks without citation
- Cite every other external source (papers, blog posts, Stack Overflow, tutorials)
- Using LLM assistants is allowed; you must state how you used them and you are responsible for every line you submit

Acknowledgments

The lecture slides and notebooks of this course build on **Andreas C. Müller's *Applied Machine Learning* course** (Columbia University, COMS W4995, Spring 2019–2020), released under the CC0 license:

- Slides and code: <https://github.com/amueller/COMS4995-s20>
- Online book: <https://amueller.github.io/aml>
- Textbook: A. C. Müller, S. Guido, *Introduction to Machine Learning with Python*, O'Reilly

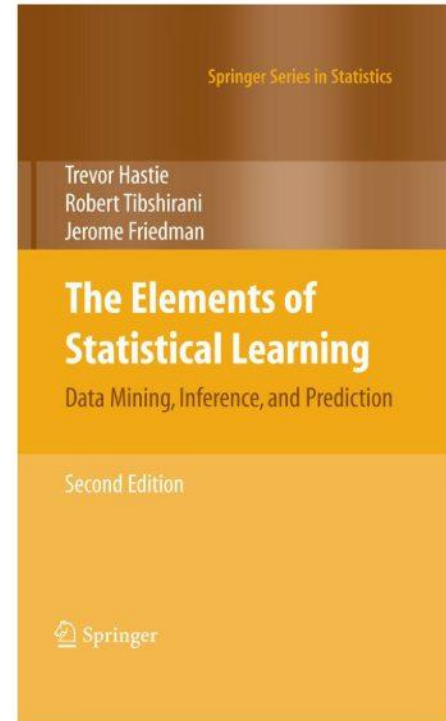
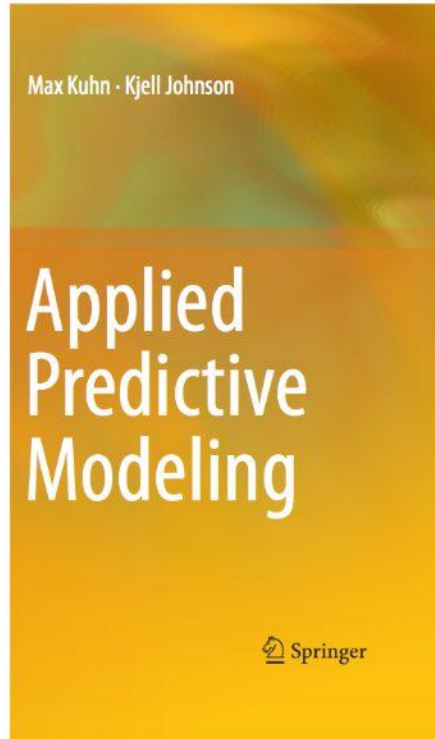
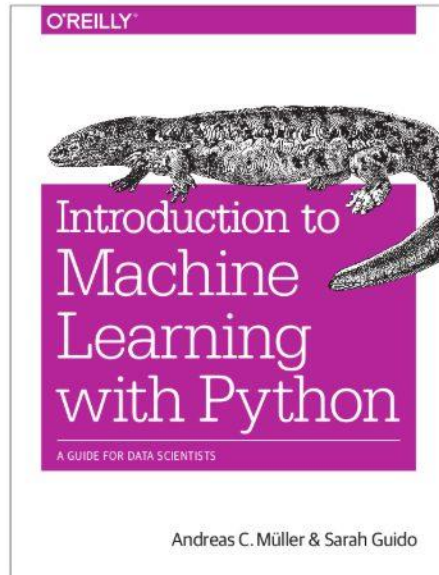
The material has been reorganized, updated to current library versions, and extended with new content (experiment tracking, the ML lifecycle, time series).

Course Tooling

- **uv**: Python environments and dependencies (uv sync)
- **Git & GitHub**: version control for code and papers
- **Jupyter**: notebooks for experiments
- **MLflow**: experiment tracking; from Week 2 on, every notebook and homework logs its runs

```
uv sync
uv run mlflow ui          # http://127.0.0.1:5000
```

Books



<http://ciml.info/>

Part 1: What and Why of Machine Learning

A motivating case: hospital readmissions

A hospital wants to know, at discharge, which patients will be **readmitted within 30 days**.

- What are x and y ? When is y known?
- Classification or regression? What is the baseline?
- What does a false negative cost? A false positive?
- What would the hospital *do* with a prediction?

What is machine learning?

Types of Machine Learning

- Supervised
- Unsupervised
- Reinforcement

Supervised Learning

$$(x_i, y_i) \sim p(x, y) \text{ i.i.d.}$$

$$x_i \in \mathbb{R}^p$$

$$y_i \in \mathbb{R}$$

$$f(x_i) \approx y_i$$

Generalization

Not only
also for new data:

$$f(x_i) \approx y_i,$$

$$f(x) \approx y$$

Unsupervised Learning

$$\mathbf{x}_i \sim p(\mathbf{x}) \text{ i.i.d.}$$

Learn about p .

- Clustering
- Dimensionality reduction
- Topic modeling

Other kinds of learning

- Semi-supervised
- Active Learning
- Forecasting
- ...

Classification and Regression

Classification

- target y discrete
- Is this patient sick?

Regression

- target y continuous
- How long will it take for the patient to recover?

Guiding Principles in Machine Learning

Goal considerations

The Cost of Complex Systems

Data driven first? yes! (or maybe)

Machine Learning first: No!

Thinking in Context!

What is the baseline?

What is the benefit?

Good and Bad Substitutes

Try it: frame your own problem

10–15 minutes, in pairs. Pick a prediction problem from your own field (a candidate for your project). Write down:

1. The unit of prediction, the features x , and the target y . When is y observed?
2. Supervised or unsupervised? Classification or regression?
3. The simplest baseline (a rule, the majority class, last year's value).
4. The metric you would report, and the quantity you actually care about. Could optimizing the metric hurt the real goal?
5. Is labeled data free or expensive? How many examples could you get?

Two pairs present in 1 minute each; the class looks for weak spots.

Communicating Results

Sidebar: Ethical Considerations



<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

Ethics: It's in the application!

Data and Data Collection

What goes wrong in practice

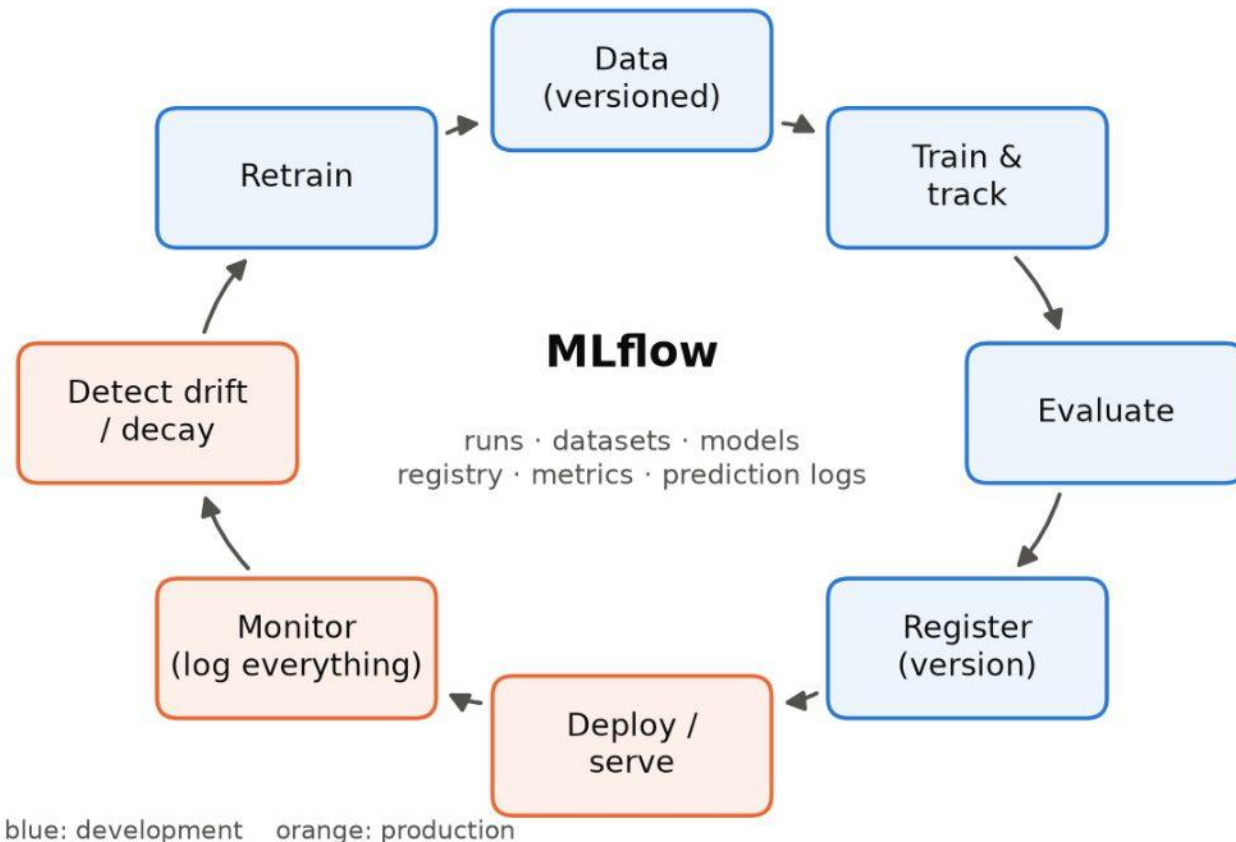
Shortcut learning. A CNN detecting pneumonia on chest X-rays scored well within the hospitals it was trained on, but partly by recognizing *which hospital* an image came from (scanner and label artifacts), since disease prevalence differed between hospitals. Performance dropped on external hospitals. Zech et al., *PLOS Medicine* 2018

A substitute that drifted. Google Flu Trends estimated flu prevalence from search queries. In the 2012–13 season it overestimated doctor visits for influenza-like illness by roughly a factor of two, as search behavior and the search engine itself changed. Lazer et al., *Science* 2014

Cornerstones of this Course

- Good software engineering practices
- Problem definition and success measures
- Feature engineering and data cleaning
- Strength and weaknesses of different algorithms
- Model selection best practices
- Experiment tracking and the ML lifecycle

The Machine Learning Lifecycle



General coding guidelines

- Don't be clever!
- Make it readable!
- Future you is the most likely person to try to understand your code.
- Avoid writing code.

Python basics

Python environments

- Don't use the system Python!
- One virtual environment per project
- Record dependencies in `pyproject.toml`, exact versions in a lock file
- We use **uv** (<https://docs.astral.sh/uv/>)

Working with uv

```
# get the course environment (reads pyproject.toml + uv.lock)
```

```
$ uv sync
```

```
# run anything inside the environment, no activation needed
```

```
$ uv run python train.py
```

```
$ uv run jupyter lab
```

```
# add a dependency (updates pyproject.toml and uv.lock)
```

```
$ uv add lightgbm
```

```
$ uv add --dev pytest pytest-cov
```

```
# start your own project
```

```
$ uv init my-project
```

Environments and Jupyter Kernels

- Environment != kernel: a kernel is the process that runs your notebook
- Simplest: start Jupyter from the project environment

```
$ uv run jupyter lab
```

- To use the environment from another Jupyter or VS Code, register a kernel:

```
$ uv run python -m ipykernel install --user --name aie683 --display-name "Python (A
```

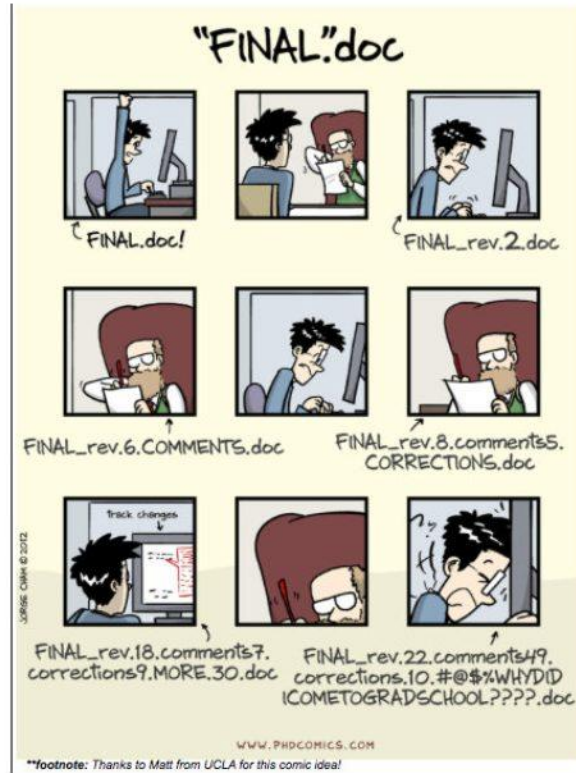
Editors and linters

- VS Code / PyCharm
- JupyterLab
- Google Colab
- Linter + formatter: `ruff` (`uvx ruff check .`)

Questions ?

Part 2: Git, GitHub and Testing

Why do I care?



Configuration

```
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"  
git config --global init.defaultBranch main  
git config --global core.editor "code --wait"
```

Show your configuration:

```
git config --list
```

Creating a repository

```
$ mkdir homework1  
$ cd homework1
```

```
$ git init
```

Initialized empty Git repository in /tmp/homework1/.git/

```
$ ls -a
```

```
.  ..  .git
```

```
$ git status
```

On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)

Cloning a repository

The screenshot shows the GitHub interface for the repository 'aml-spring-19 / homework-1'. At the top, there are buttons for 'Unwatch', 'Star' (1), and 'Fork' (0). Below this is a navigation bar with 'Code', 'Issues' (0), 'Pull requests' (0), 'Projects' (0), 'Wiki', 'More', and 'Settings'. The main content area shows 'Base Repository for Homework 1' with an 'Edit' button. Below this, it says 'Manage topics'. A summary bar indicates '1 commit', '1 branch', '0 releases', and '1 contributor'. A dropdown menu shows the current branch is 'master'. Below this, a list of files is shown: '.gitignore' and 'README.md', both marked as 'Initial commit'. A modal window is open on the right, titled 'Clone with SSH', with a link to 'Use HTTPS'. It instructs the user to 'Use an SSH key and passphrase from account.' and shows the repository URL 'git@github.com:aml-spring-19/homework' with a copy icon. At the bottom of the modal is a 'Download ZIP' button.

aml-spring-19 / homework-1

Unwatch 1 Star 1 Fork 0

Code Issues 0 Pull requests 0 Projects 0 Wiki More Settings

Base Repository for Homework 1 Edit

Manage topics

1 commit 1 branch 0 releases 1 contributor

Branch: master

Create new file Find file Clone or download

Clone with SSH Use HTTPS

Use an SSH key and passphrase from account.

git@github.com:aml-spring-19/homework

Download ZIP

amuel Initial commit

.gitignore Initial commit

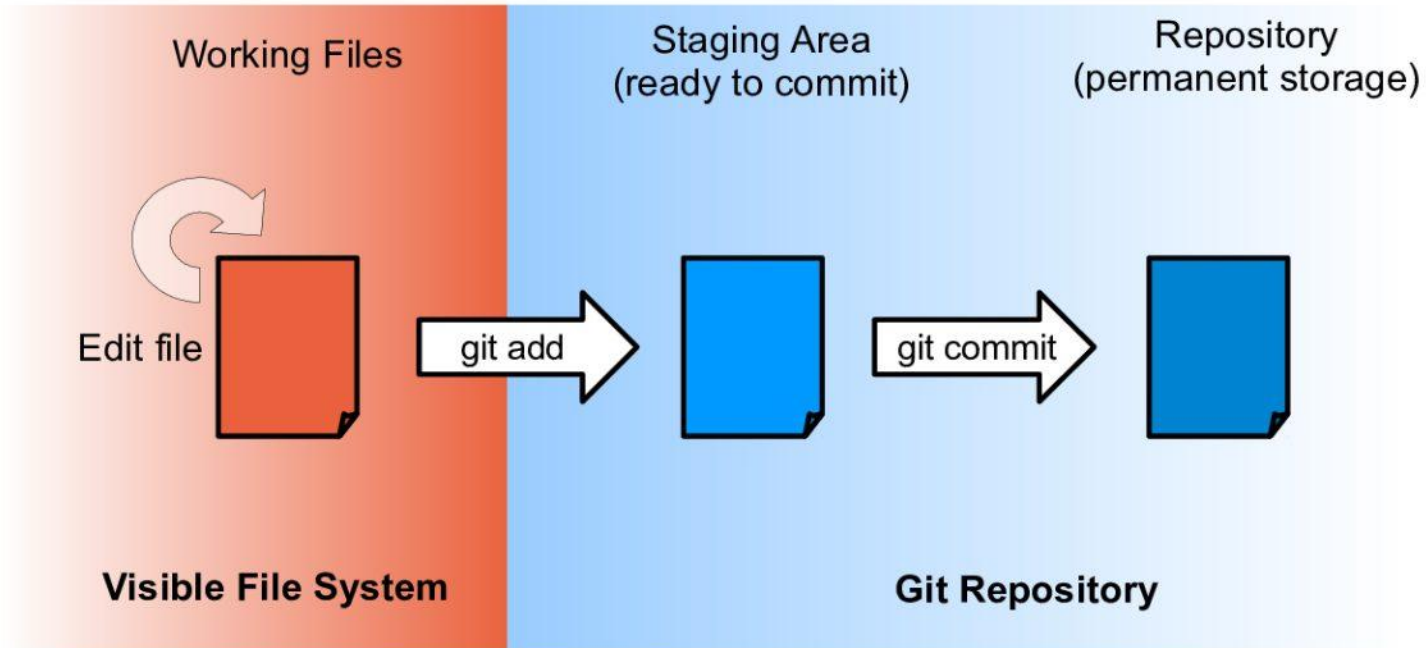
README.md Initial commit

homework-1

Base Repository for Homework 1

```
$ git clone git@github.com:<user>/<repo>.git
```

Workflow?



Changing a File

```
$ echo "print('Hello world!')" >> task1.py
```

```
$ git status
```

On branch main

No commits yet

Untracked files:

(use "git add <file>..." to include in what will be committed)

task1.py

nothing added to commit but untracked files present (use "git add" to track)

```
$ git add task1.py
```

Viewing your history

```
$ git commit -m "say hello"
```

```
[main (root-commit) 7d139fb] say hello  
1 file changed, 1 insertion(+)  
create mode 100644 task1.py
```

```
$ git log
```

```
commit 7d139fb317ecfa7d629654b709747e91ececaca444 (HEAD -> main)  
Author: Your Name <you@example.com>  
Date: Thu Oct 1 09:12:29 2026 +0300
```

```
say hello
```

A note on viewing changes

Changes between working directory and what was last staged

```
git diff
```

Changes between staging area and last commit

```
git diff --staged
```

Recovering Older Versions of Files

- Overwrite task1.py:

```
$ echo "print('goodbye, cruel world!')" > task1.py  
$ cat task1.py
```

```
print('goodbye, cruel world!')
```

Recover last recorded version:

```
$ git restore --source=HEAD task1.py  
$ cat task1.py
```

```
print('Hello world!')
```

Branches

```
$ git switch -c new_feature
```

Switched to a new branch 'new_feature'

make some changes, add, commit...

Moving between branches:

```
$ git switch main
```

Switched to branch 'main'

changes are not present in main..

Merge

- Fast-forward merge:

```
* 6cec4ed (HEAD -> another_one) E
* 1e96a3b D
* 513ced1 (master) C
* b5ba00a B
* db928a0 A

/tmp/git_graphs [git::master] [andy@dsi-amueller] [15:38]
> git merge another_one
Updating 513ced1..6cec4ed
Fast-forward
 D | 0
 E | 0
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 D
 create mode 100644 E

* 6cec4ed (HEAD -> master, another_one) E
* 1e96a3b D
* 513ced1 C
* b5ba00a B
* db928a0 A
```

- Merge-commits:

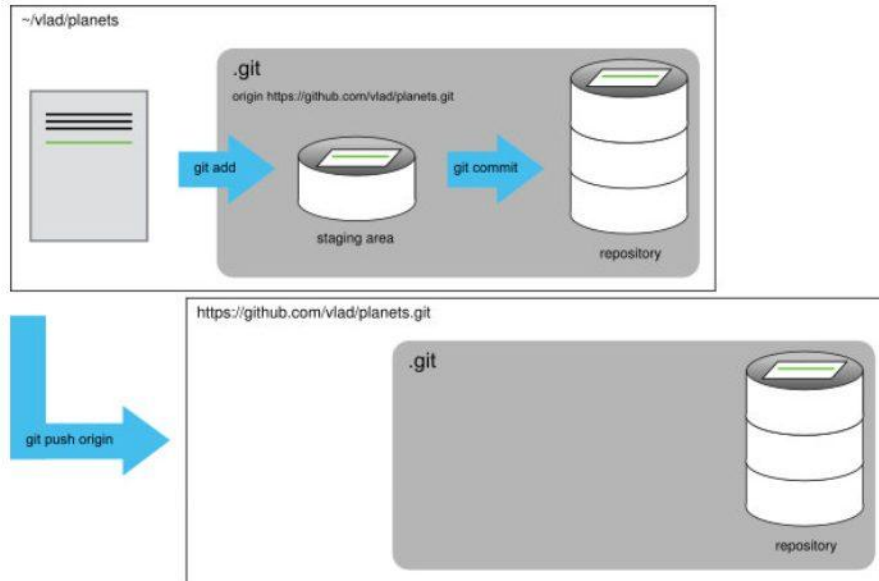
```
* 43563a5 (HEAD -> master) G
* c3ea8c8 F
| * 6cec4ed (another_one) E
| * 1e96a3b D
|/
* 513ced1 C
* b5ba00a B
* db928a0 A

/tmp/git_graphs [git::master] [andy@dsi-amueller] [15:43]
> git merge another_one
Merge made by the 'recursive' strategy.
 D | 0
 E | 0
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 D
 create mode 100644 E

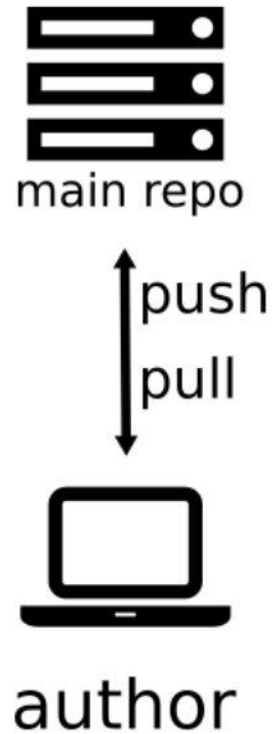
* d6fedb0 (HEAD -> master) Merge branch 'another_one'
| \
| * 6cec4ed (another_one) E
| * 1e96a3b D
| * | 43563a5 G
| * | c3ea8c8 F
|/
* 513ced1 C
* b5ba00a B
* db928a0 A
```

Remote repository

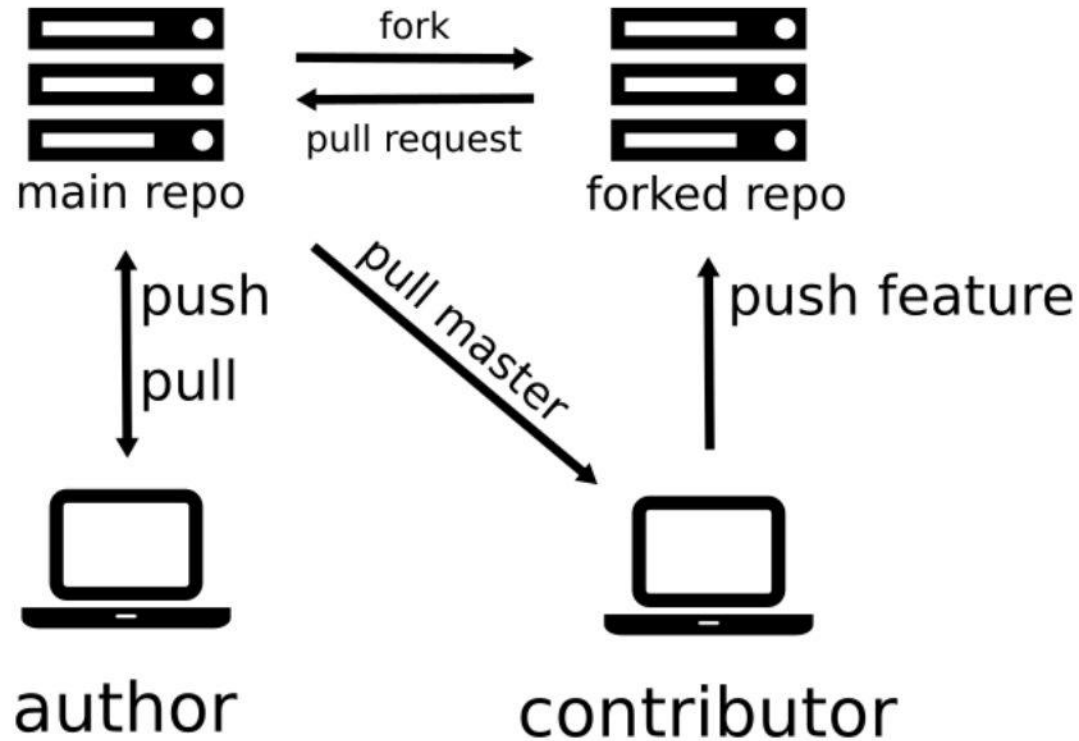
- central location everyone can see
- requires network ?
- GitHub, GitLab, Bitbucket
- public vs private



GitHub pull request workflow



GitHub pull request workflow



Typical Workflow

- Clone
- Branch
- Add, commit, add, commit, add, commit, ...
- Merge / rebase
- Push

.gitignore

```
# Python / uv
__pycache__/
.venv/

# Jupyter
.ipynb_checkpoints/

# experiment tracking and outputs
mlruns/
mlartifacts/
outputs/

# data: download at runtime, don't commit
data/
.csv
.parquet
```

Unit Tests and integration tests

Why test?

- Ensure that code works correctly.
- Ensure that changes don't break anything.
- Ensure that bugs are not reintroduced.
- Ensure robustness to user errors.
- Ensure code is reachable.

How to test?

- pytest – <https://docs.pytest.org>
- Searches for all `test_*.py` files, runs all `test_*` functions.
- Reports nice errors!
- Run it in the project environment:

```
$ uv add --dev pytest pytest-cov  
$ uv run pytest
```

Example

```
# content of inc.py
```

```
def inc(x):  
    return x + 2
```

```
# content of test_sample.py
```

```
from inc import inc
```

```
def test_answer():  
    assert inc(3) == 4
```

```
> py.test test_sample.py
```

```
===== test session starts =====
```

```
platform linux -- Python 3.5.2, pytest-2.9.2, py-1.4.31, pluggy-0.3.1
```

```
rootdir: /tmp, inifile:
```

```
plugins: cov-2.4.0, timeout-1.2.0
```

```
collected 1 items
```

```
test_sample.py F
```

```
===== FAILURES =====
```

```
_____ test_answer _____
```

```
def test_answer():  
>     assert inc(3) == 4  
E       assert 5 == 4  
E         + where 5 = inc(3)
```

```
test_sample.py:7: AssertionError
```

```
===== 1 failed in 0.01 seconds =====
```

Testing ML code

```
# test_model.py
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression

def test_beats_chance():
    X, y = make_classification(n_samples=200, random_state=0)
    model = LogisticRegression().fit(X[:150], y[:150])
    assert model.score(X[150:], y[150:]) > 0.6

def test_predict_shape():
    X, y = make_classification(n_samples=50, random_state=0)
    pred = LogisticRegression().fit(X, y).predict(X)
    assert pred.shape == y.shape
```

Continuous integration (with GitHub)

What is Continuous integration?

- Run command on each commit (or each PR).
- Unit testing and integration testing.
- Can act as a build-farm (for binaries or documentation).
- Requires clear declaration of dependencies.
- Build matrix: Can run on many environments.
- Common services: GitHub Actions, GitLab CI, CircleCI, Azure Pipelines

Setting up GitHub Actions

File `.github/workflows/tests.yml`:

```
name: tests
on: [push, pull_request]

jobs:
  test:
    runs-on: ubuntu-latest
    strategy:
      matrix:
        python-version: ["3.11", "3.12"]
    steps:
      - uses: actions/checkout@v4
      - uses: astral-sh/setup-uv@v6
        with:
          python-version: ${ matrix.python-version }
      - run: uv sync --locked
      - run: uv run pytest
```

Try it: a tested, versioned project

15 minutes, on your laptop.

```
$ uv init aie683-sandbox && cd aie683-sandbox  
$ uv add scikit-learn && uv add --dev pytest
```

1. Write `model.py` with a function `fit_model(X, y)` and a test `tests/test_model.py` that checks it beats the majority class on `load_breast_cancer()`.
2. `uv run pytest`, then `git add`, `git commit`.
3. Break the function, commit, then recover the working version with `git log` and `git restore` (or `git reset`).
4. Push to a new private GitHub repository.

Bonus: add the GitHub Actions workflow from the previous slides and watch the check turn green.

Questions ?

Backup

Andreas
Home

Andreas Mueller

News Feed

Messenger

SHORTCUTS

Amazon Machine L... 4

Freunde +1

EXPLORE

Events

Pages

Groups

Friend Lists

Pokes

Photos

On This Day

Suggest Edits

Offers

Moments 20+

See More...

CREATE

Ad

Page

Group

Event

Fundraiser

untapt

Like Page

Sponsored

Accomplish more in 2017. See this year's most coveted fintech engineering jobs.

add a to your resume.

New job. New adventure.

We've trained computer models on thousands of successful job applications to understand how the best engineers land their dream jobs. Now it's time to land yours.

UNTAPT.COM

Learn More

4

3 Comments

Like

Comment

Share

Juliana Vanderlee

is watching Tom Price's confirmation hearing.

6 mins ·

All of these people are literally some of the worst possible choices for their positions.

Like

Comment

Share

2

Write a comment...

Cédric Archambeau and 1 other

TRENDING

Gmail
190K people talking about this

Pierce Brosnan
9.8K people talking about this

Wii U
120K people talking about this

See More

CONNECT WITH FACEBOOK

Internet.org by Facebook

7,615,868 likes

Instagram

40,973,994 likes

Facebook Engineering

8,548,617 likes

Mark Zuckerberg

84,629,591 followers

PEOPLE YOU MAY KNOW

See All

Dmytro Pavlchenko

4 mutual friends

Add Friend

Lee Tanenbaum

2 mutual friends

Add Friend

English (US)

Español

Português (Brasil)

Français (France)

Deutsch

+

Privacy · Terms · Advertising · Ad Choices · Cookies · More

Facebook © 2017

68 / 122

Andreas Home

Andreas Mueller added 8 new photos.
 June 6 at 5:07pm

Bye bye hong kong

Like · Comment · Share

Single, Phil, Clara, Stephanie, Wai-Ching and 6 others like this.

June 6 at 7:39pm · Like

Write a comment...

SPONSORED

Da ist Abwechslung drin!
 franziskaner-weissbier.de

 3 Sorten Franziskaner Weissbier in einem Pack. Hier Klicken und 1 von 100 Packs gewinnen!

Singles auf Facebook

 Schau dir Dating-Profil von Singles in deiner Nähe an.

Online Essen bestellen
 lieferheld.de

 Neu mit Lieferheld: PLZ eingeben, Lieferservice finden und genießen!

Globaler Chauffeurservice
 blacklane.com

 Fahren Sie eine Klasse besser zu günstigen Preisen – Blacklane ist weltweit verfügbar!
 28,723 people like this

Sanssouci...
 Geld vom Staat zurück!

 Steuererklärung preiswert für Arbeitnehmer, Azubis, Arbeitsuchende, Rentner, Pensionäre etc.
 Like Page · 4 people like this page

Der neue WhatsApp Tarif!
 eplus.de

 WhatsApp SIM – der revolutionäre Prepaid Tarif

English (US) · Privacy · Terms · Cookies · More · Facebook © 2014

www.amazon.com/s/ref=nb_sb_noss_1?url=search-alias%3Daps&field-keywords=machine+learning&prefix=machine+%2Caps&rh= DuckDuckGo

amazon try prime [Andrew's Amazon.com](#) [Today's Deals](#) [Gift Cards](#) [Sell](#) [Help](#) **Father's Day** is June 15 Reminded by DuckDuckGo [+ \\$2.00 / day](#)

Shop by **Department** Search [Hello, Andrew](#) [Your Account](#) [Try Prime](#) [Cart](#) [Wish List](#)

1-16 of 21,540 results for "machine learning"

Choose a Department to sort

Show results for:

Books

- Machine Learning
- Computers & Technology
- Computer Science
- Reference
- Programming Algorithms
- + See more

Kindle Store

- Computers & Technology
- Computer Programming
- + See more

+ See All 30 Departments

Refine by

Eligible for Free Shipping
Free Shipping by Amazon

Book Series

- ☐ Information Science and Statistics
- ☐ Adaptive Computation and Machine Learning series
- ☐ I Can Do It
- ☐ Use It!

Book Language

- ☐ English

Book Format

- ☐ Hardcover
- ☐ Paperback
- ☐ Kindle Edition
- ☐ HTML

Avg. Customer Review

- ☐ ★★★★★ & Up
- ☐ ★★★★☆ & Up
- ☐ ★★★☆☆ & Up
- ☐ ★★★★★ & Up

Machine Learning Resources
Find tips and tricks with these featured titles on machine learning, algorithms, sensors, and more. [Learn more](#)

Related Searches: [adaptive intelligence](#) [data science](#) [machine learning python](#)

Practical Machine Learning: Innovations in Recommendation by Ted Dunning and Elen Friedman (Apr 17, 2014)
\$0.00 Kindle Edition
Auto-delivered wholesale
★☆☆☆☆ (1)
[Books: See all 18,835 items](#)

Machine Learning: The Art and Science of Algorithms that Make Sense of Data by Peter Flach (Nov 12, 2012)
was \$54.96 **\$34.96** Paperback [+ Prime](#)
Only 16 left in stock - order soon
More Buying Choices - Paperback
\$46.96 new (40 offers)
\$35.00 used (18 offers)
★☆☆☆☆ (11)
FREE Shipping
Trade in eligible for an Amazon gift card
Other Formats: Hardcover
[Books: See all 16,820 items](#)

Understanding Machine Learning: From Theory to Algorithms by Shai Shalev-Shwartz and Shai Ben-David (May 19, 2014)
was \$60.00 **\$50.92** Hardcover [+ Prime](#)
Order in the next 11 hours and get it by Tuesday, Jun 18
\$34.35 Kindle Edition
Auto-delivered wholesale
More Buying Choices - Hardcover
\$50.92 new (8 offers)
\$52.02 used (4 offers)
FREE Shipping
Trade in eligible for an Amazon gift card
[Books: See all 15,305 items](#)

Learning From Data by Yaser S. Abu-Mostafa, Malik Magdon-Esmail and Hsuan-Tien Lin (Mar 27, 2012)
★☆☆☆☆ (96)
\$28.00 new (3 offers)
\$40.00 used (13 offers)
[Books: See all 16,325 items](#)

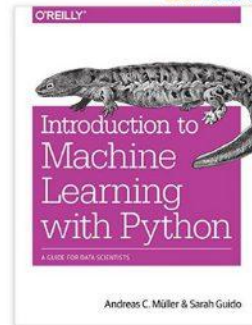
www.amazon.com/Fathers-Day-Gifts-Sale/?ref=

Introduction to Machine Learning with Python: A Guide for Data Scientists 1st Edition

by Andreas C. Müller (Author), Sarah Guido (Author)

★★★★☆ 9 customer reviews

Look inside



ISBN-13: 978-1449369415

ISBN-10: 1449369413

[Why is ISBN important?](#)

Have one to sell?

[Sell on Amazon](#)

Add to List

Share [✉](#) [f](#) [t](#) [p](#)

Kindle \$24.99

Paperback
\$41.23

Other Sellers
from \$26.53

Buy new

In Stock.

Ships from and sold by Amazon.com. Gift-wrap available.

[Prime](#)

Want it tomorrow, Jan. 19? Order within 5 hrs 20 mins and choose One-Day Shipping at checkout. [Details](#)

[Prime](#) \$41.23

List Price: \$49.99 Save: \$8.76 (18%)

36 New from \$26.53

Qty: 1

[Add to Cart](#)

[Turn on 1-Click ordering](#)

Ship to:

Andreas C Mueller- New York - 10009

More Buying Choices

36 New from \$26.53 | 15 Used from \$29.09

51 used & new from \$26.53

[See All Buying Options](#)

Machine learning has become an integral part of many commercial applications and research projects, but this field is not exclusive to large companies with extensive research teams. If you use Python, even as a beginner, this book will teach you practical ways to build your own machine learning solutions. With all the data available today, machine learning applications are limited only by your imagination.

You'll learn the steps necessary to create a successful machine-learning application with Python and the [Read more](#)

[Report incorrect product information.](#)



Save up to 90%
on textbooks

[Shop now](#)

Frequently Bought Together

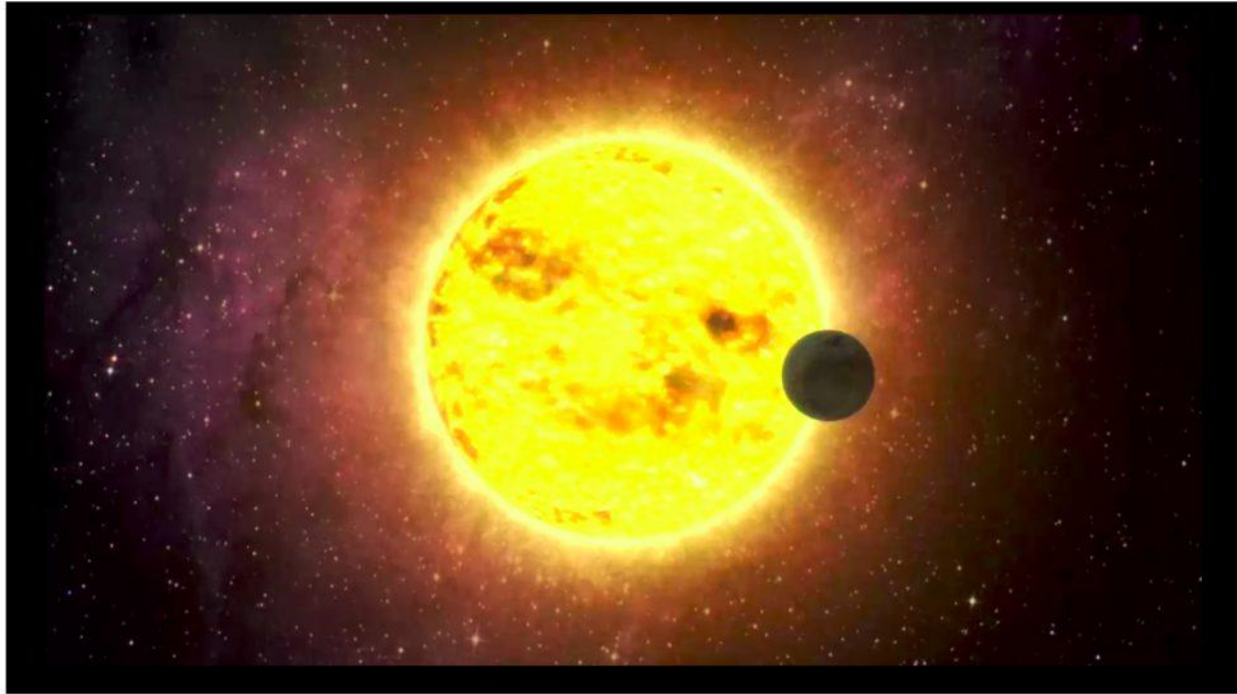


Total price: \$108.29

[Add all three to Cart](#)

[Add all three to List](#)

Science!

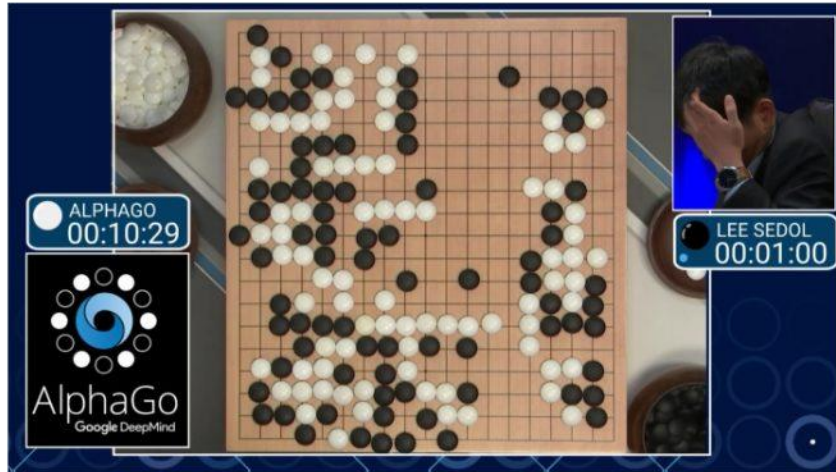


Types of Machine Learning

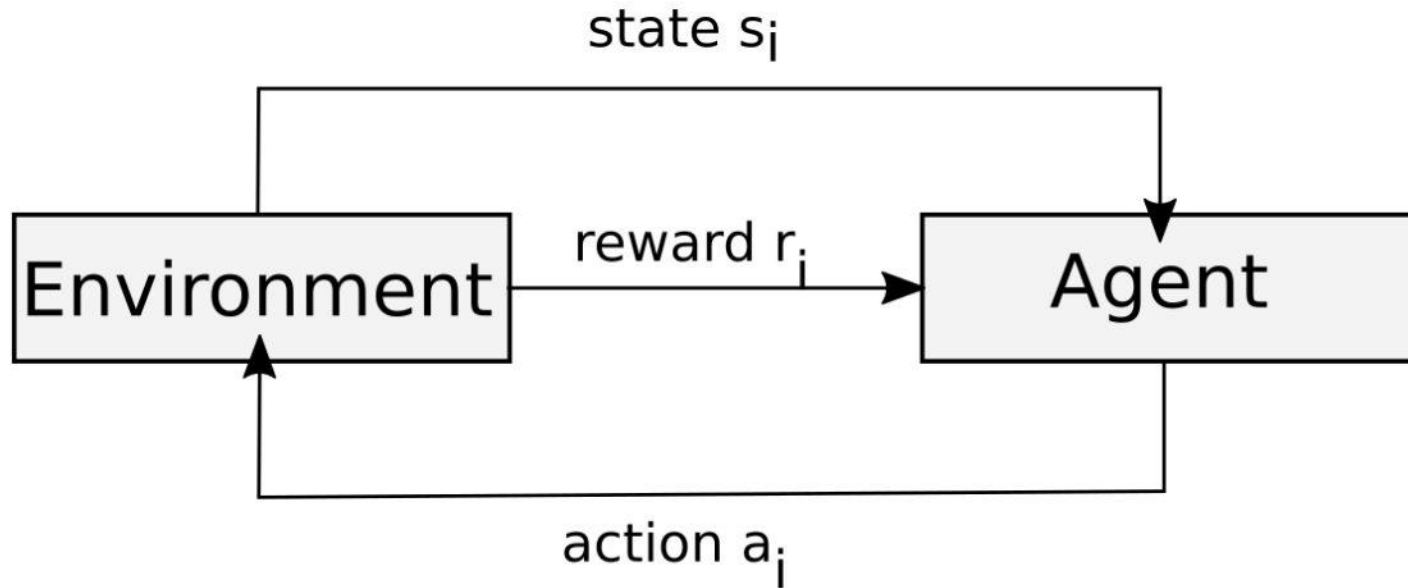
Examples of Supervised Learning

- spam detection
- medical diagnosis
- ad click prediction

Reinforcement Learning



Explore & Learn





Relationship to Statistics

Statistics

- model first
- inference emphasis

Machine learning

- data first
- prediction emphasis

Relationship to Statistics

Statistics

- model first
- inference emphasis

Machine learning

- data first
- prediction emphasis

Explainable Results

These recommendations are based on items you own and more.

view: [All](#) | [New Releases](#) | [Coming Soon](#)

1.



R for Data Science: Import, Tidy, Transform, Visualize, and Model Data

by Hadley Wickham (January 5, 2017)

Average Customer Review: ★★★★★ (4)

In Stock

List Price: ~~\$39.99~~

Price: \$32.91

[28 used & new](#) from \$28.91

[Add to Cart](#) [Add to Wish List](#)

☐ I own it ☐ Not interested ☒ ★★★★★ Rate this item

Recommended because you purchased **Data Science from Scratch: First Principles with Python** and more ([Fix this](#))

2.



Storytelling with Data: A Data Visualization Guide for Business Professionals

by Cole Nussbaumer Knaflic (November 2, 2015)

Average Customer Review: ★★★★★ (137)

In Stock

List Price: ~~\$39.95~~

Price: \$29.71

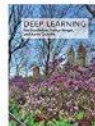
[94 used & new](#) from \$20.00

[Add to Cart](#) [Add to Wish List](#)

☐ I own it ☐ Not interested ☒ ★★★★★ Rate this item

Recommended because you purchased **Mindset: The New Psychology of Success** and more ([Fix this](#))

3.



Deep Learning (Adaptive Computation and Machine Learning series)

by Ian Goodfellow (November 18, 2016)

Average Customer Review: ★★★★★ (20)

In Stock

List Price: ~~\$89.00~~

Price: \$68.34

[18 used & new](#) from \$68.34

[Add to Cart](#) [Add to Wish List](#)

☐ I own it ☐ Not interested ☒ ★★★★★ Rate this item

Recommended because you purchased **Data Science from Scratch: First Principles with Python** and more ([Fix this](#))

Free vs Expensive Data

Free

Expensive

Free vs Expensive Data

Free

Predict observable events

- Stock market
- Clicks
- House numbers

Expensive

Automate complex process

- Diagnosis
- Drug Trial
- Chip Design

The cost (and benefit?) of BigData
Subsample to RAM (cloud machines:
terabytes)

Programs must be written for people to read, and only incidentally for machines to execute.

Harold Abelson (wizard book)

Everyone knows that debugging is twice as hard as writing a program in the first place. So if you're as clever as you can be when you write it, how will you ever debug it?

Brian Kernighan

Why Python?

- General purpose language
- Great libraries
- Easy to learn / use
- Contenders: R (Julia?)

The two language problem

Python is sloooow...

- Numpy: C
- Scipy: C, Fortran
- Pandas: Cython, C, Python
- Scikit-learn: Cython, Python
- CPython: C

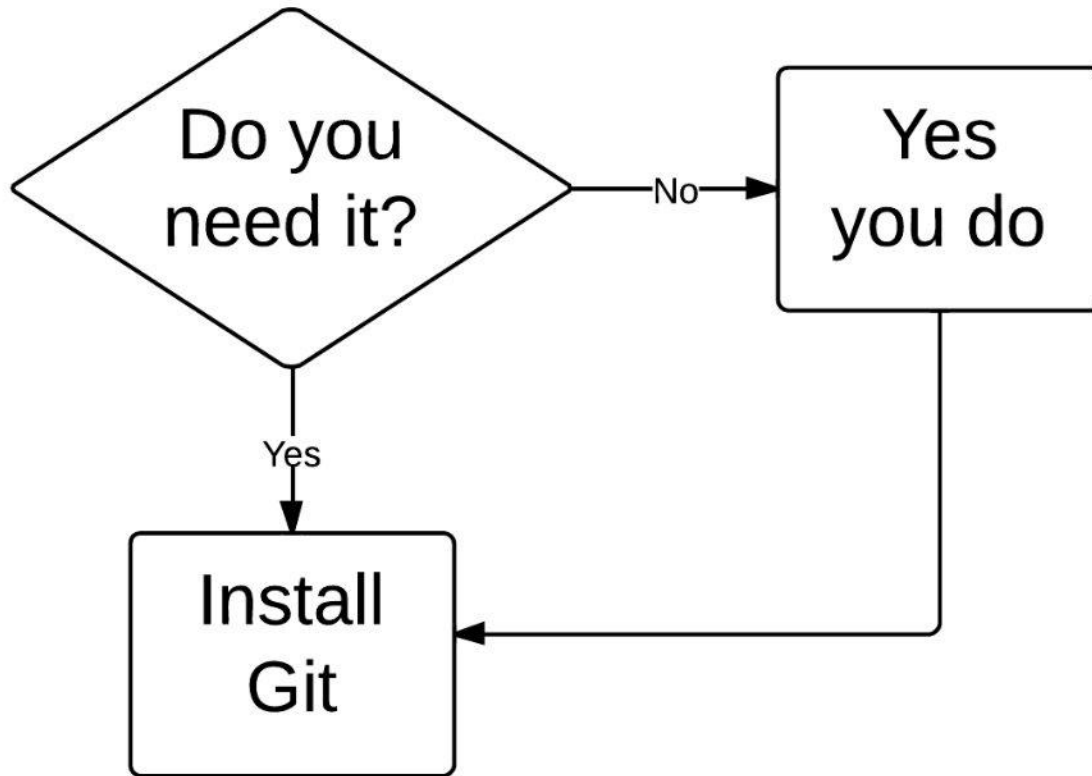
Python versions

- Python 2 is dead (end of life since 2020)
- This course: **Python 3.12**, pinned in `pyproject.toml`
- Supported versions: see <https://devguide.python.org/versions/>

Dynamically typed, interpreted

- Invalid code lying around
- Code is less self-documenting

Version Control Flowchart



Git and GitHub

<https://docs.github.com/en/get-started>

<https://rogerdudler.github.io/git-guide/>

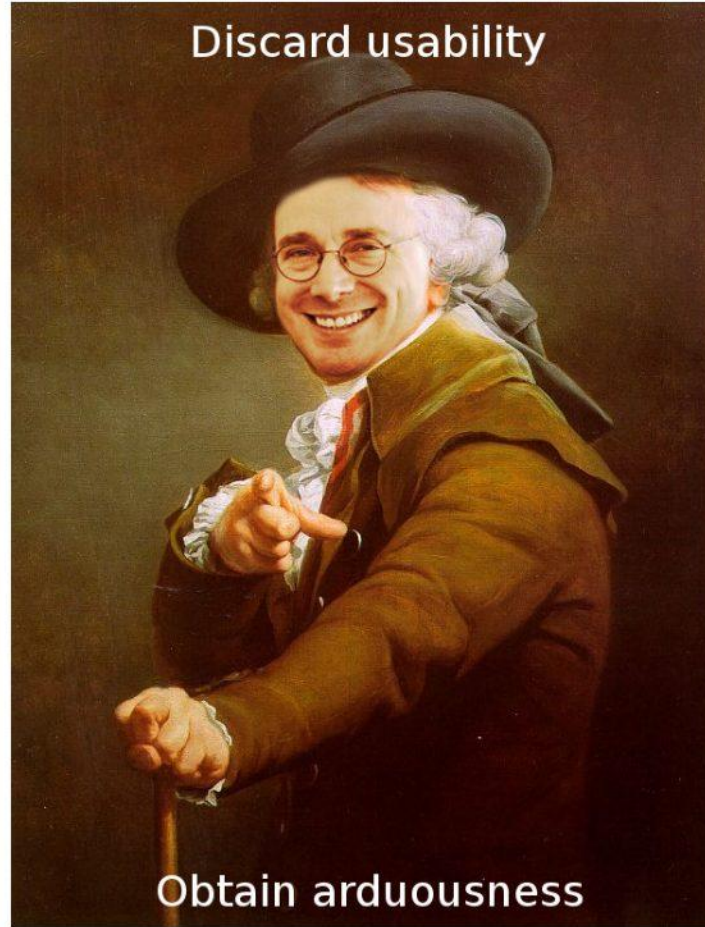
THIS IS GIT. IT TRACKS COLLABORATIVE WORK
ON PROJECTS THROUGH A BEAUTIFUL
DISTRIBUTED GRAPH THEORY TREE MODEL.

COOL. HOW DO WE USE IT?

NO IDEA. JUST MEMORIZE THESE SHELL
COMMANDS AND TYPE THEM TO SYNC UP.
IF YOU GET ERRORS, SAVE YOUR WORK
ELSEWHERE, DELETE THE PROJECT,
AND DOWNLOAD A FRESH COPY.



Discard usability



Obtain arduousness

Referencing different versions

- Shorthand for different versions of a repository (refers to commits)
 - Current Version (most recent commit): HEAD
 - Version before current: HEAD~1
 - Version before that: HEAD~2
- Each of these also has a commit hash
 - use `git log` to get appropriate hash

Exploring History

Changes made in the last commit

```
git diff HEAD~1
```

Changes made in the last 2 commits

```
git diff HEAD~2
```

Changes made since commit hash...

```
git diff 0b0d55e
```

Recovering Older Versions of whole repo

```
$ echo "print('goodbye, cruel world!')" > task1.py  
$ git add task1.py  
$ git commit -m "saying goodbye"
```

```
[main 95f3b39] saying goodbye  
1 file changed, 1 insertion(+), 1 deletion(-)
```

```
$ git log --oneline
```

```
95f3b39 (HEAD -> main) saying goodbye  
7d139fb say hello
```

Resetting:

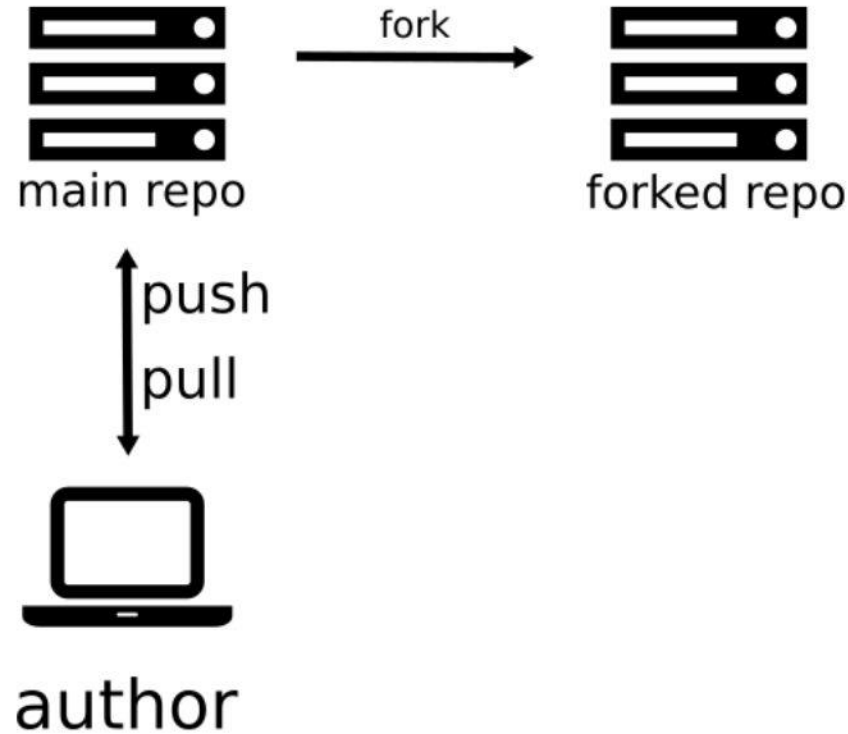
```
$ git reset --hard HEAD~1
```

HEAD is now at 7d139fb say hello

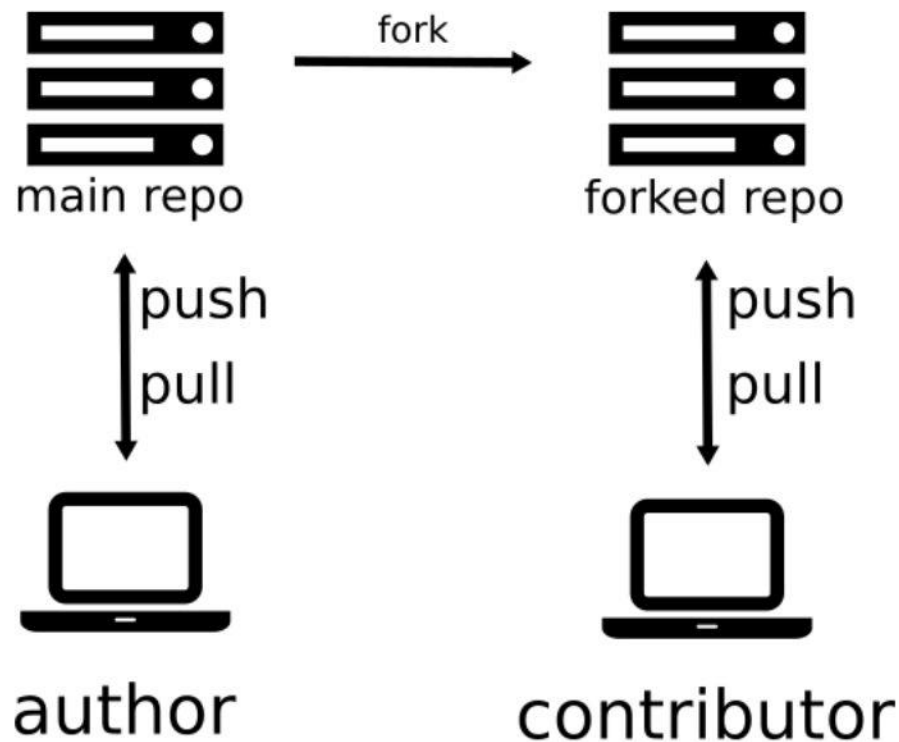
Collaborating using Git & GitHub

GitHub

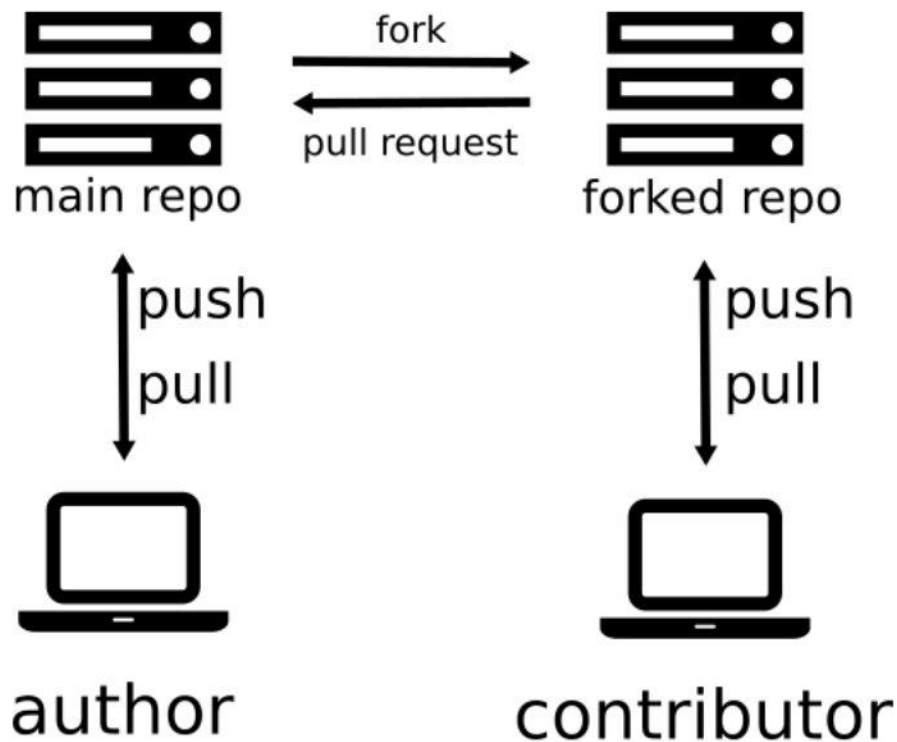
GitHub pull request workflow



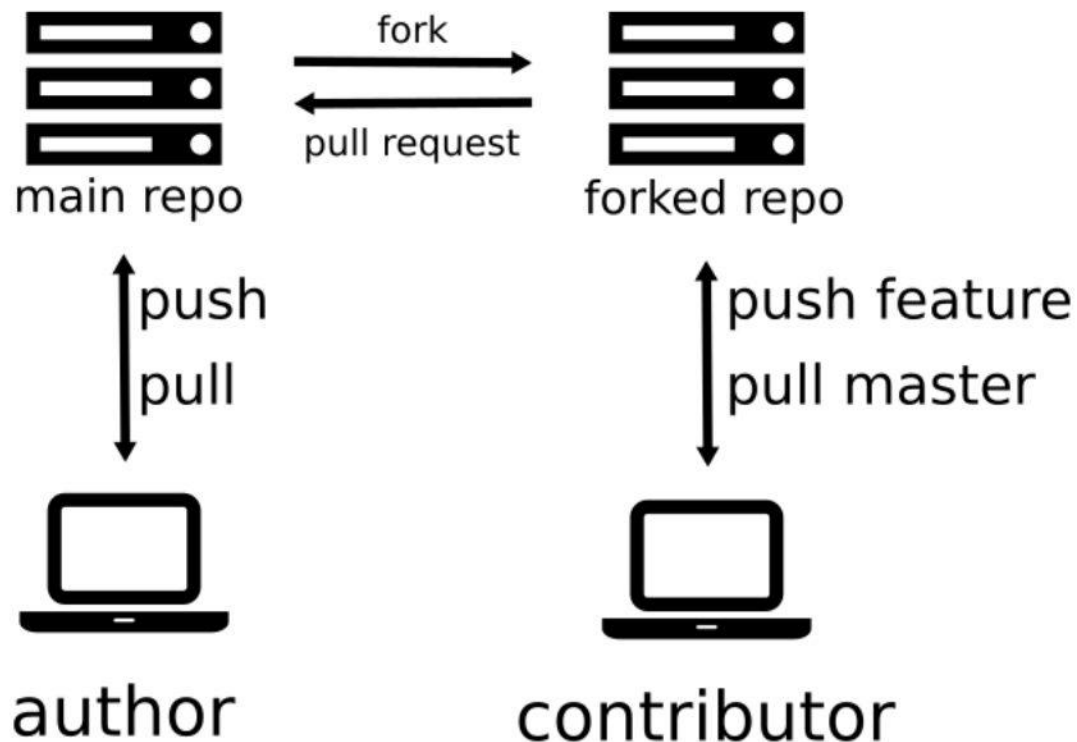
GitHub pull request workflow



GitHub pull request workflow



GitHub pull request workflow



Some tips

- `$ git status`
- Install shell plugins for status and branch (`oh-my-zsh`)

```
/home/andy/checkout/scikit-learn [git::master *] [andy@dsi-amueller] [16:35]  
> █
```

- Set editor, pager and diff-tool (check out meld!)
- Use `.gitignore`

Git log

```
/home/andy/checkout/scikit-learn [git::master *] (andy@dsi-amaeller) [16:43]
> git log --oneline --decorate --all --graph -n 50
* 9616act (HEAD -> master, upstream/master) CI remove obsolete comment
* 7978119 [MRG] #8218: in FAQ, link deep learning question to GPU question (#8220)
* be305ce TST/FIX Add check for estimator: parameters not modified by 'fit' (#7846)
* aaebee1 FIX Issue #8173 - pass n_neighbors in MI computation (#8181)
* 4826883 Call sorted on lfw folder path contents (#7648)
* 4907029 [MRG+3] FIX Memory leak in MAE; Use safe realloc; Acquire GIL only when raising; Propagate all errors to python interpreter level (#7811) (#8002)
* 08772c4 FIX Ensure coef_ is an ndarray when fitting LassoLars (#8160)
* 5d6460d MNT/BLD Use GitHub's merge refs to test PRs on CircleCI (#8211)
* 66443aa [MRG+1] Add prominent mention of Laplacian Eigenmaps (#8155)
* bddda7b [MRG + 2] [MAINT] Update to Sphinx-Gallery 0.1.7 (#7986)
* aea6462 [MRG+1] Fixes #8198 - error in datasets.make moons (#8199)
* 0eb33ad TRAVIS fix flake8 diff.sh check files (#8208)
* eabaef3 DOC add missing parentheses in TfIdfTransformer docstring
* 3dc8d2f DOC additional fixes to 20 newsgroups to prevent TypeError (#8204)
* cbddb92 removed stray space in '__main__' (#8203)
* ca687ba Upgrade html documentation to jQuery v3.1.1 (#8145)
* 84cc67b Clarify error message for min_samples_split. (#8167)
* 2a1408a fixing typo in cs_mse_path deprecation (#8176)
* fd84a56 [MRG + 1] Fix the cross_val_predict function for method='predict_proba' (#7889)
* 4910e11 DOC Fix link (#8171)
* 8998856 Fix Ridge floating point instability (#8154)
* 2f7f5a1 [MRG + 1] Add fowlkess-mallows and other supervised cluster metrics to SCORERS dict so it can be used in hyper-param search (#8117)
* 8695ff5 [MRG + 1] add partial fit to multioutput module (#8054)
* 0b02125 [MRG] FIX Avoid default mutable argument in constructor of AgglomerativeClustering (#8153)
* d0ce0d9 [MRG+2] Avoid failure in first iteration of RANSAC regression (#7914)
* e874398 [MRG+1] DOC: complete list of online learners (#8152)
* e0c60fe FIX sphinx gallery rendering of plot digits pipe example
* 84349a7 [MRG+1] Deprecate ridge.alpha param on SparsePCA.transform() (#8137)
* c49ced9 DOC: updating GridSearchCV's n_jobs parameter (#8106)
* 2cb7e47 [MRG+1] fowlkess mallows score: more unit tests (Fixes #8101) (#8140)
* 543b056 [MRG+1] Add DBSCAN support for additional metric params (#8139)
* d7e77ce [MRG+1] Fix "cite us" link in sidebar (#8142)
* 288827b [MRG] update copyright years for 2017 (#8138)
* e2adb77 DOC Fix typo in FAQ (#8132)
* 980a49b FIX Split data using _safe_split in _permutaion test score (#5697)
* ab1c4d4 [MRG+1] Catch cases for different class size in MLPClassifier with warm start (#7976) (#8035)
* dcf24b9 (origin/repr_give_up, repr_give_up) i have no idea what I'm doing
* b25fa4b pep8
* 8b30325 playing around, then giving up
* 0d1398a (upstream/ignore_lambda_to_diff_errors) MNT Ignore E731: Use a def instead of lambda
/ /
* d97d13e DOC add sklearn-crfsuite to related projects (#7878)
* fcb706a [MRG+3] Fused types for MultiTaskElasticNet (#8061)
* 096a9cb [MRG] MAINT Python 3.6 fixes (#8123)
* e088685 DOC Fix indentation errors and username links (#8121)
* 4f3c60c [MRG+2] FIX IsolationForest(max_features=0.8).predict(X) fails input validation (#5757)
```

**I DON'T DO GIT LOG ALL THE TIME,
BUT WHEN I DO JUST REMEMBER**



**(A DOG) --ALL --DECORATE --ONELINE --
-GRAPH**

memegenerator.net

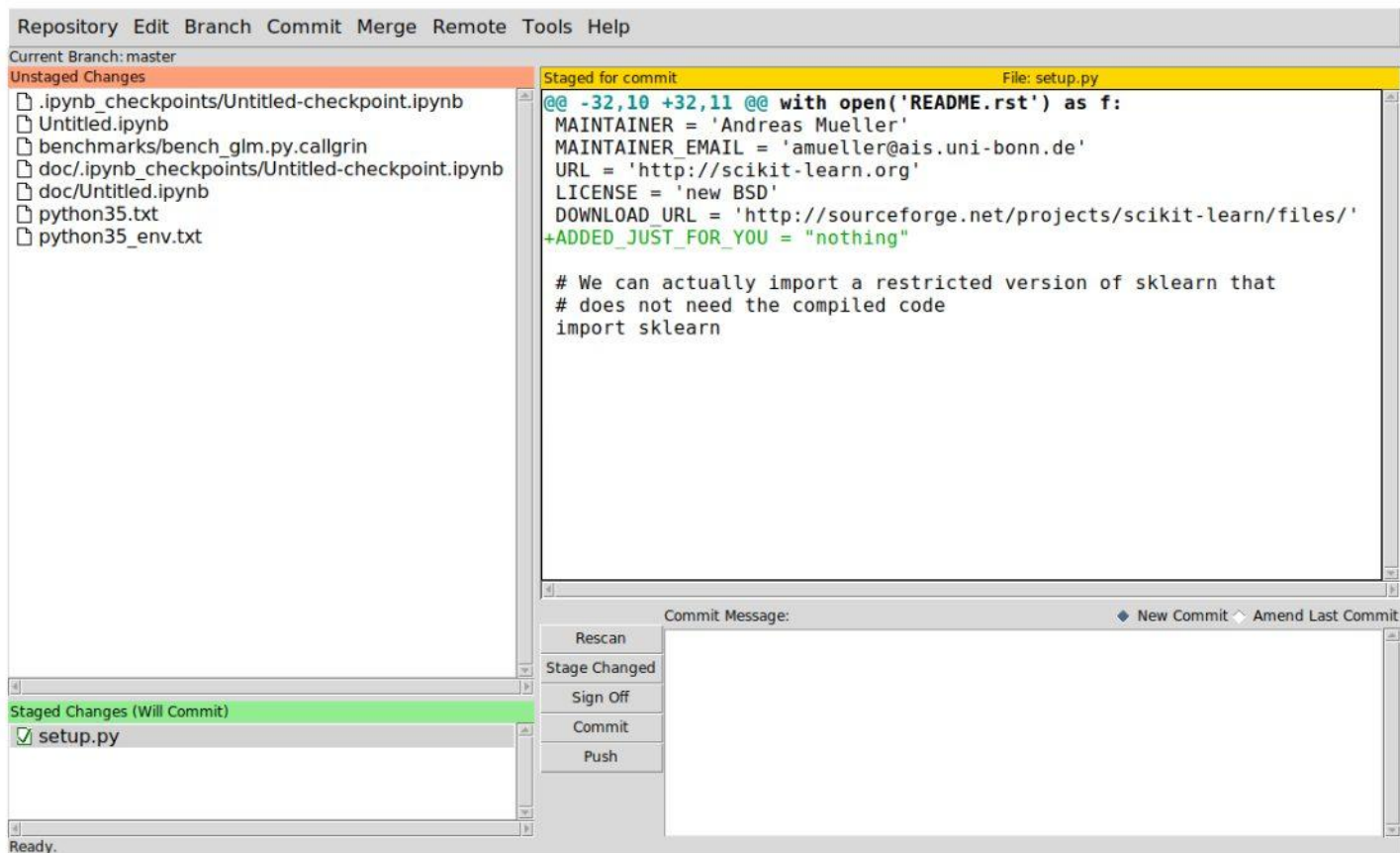
Understanding Git

- Working directory
- Repository (Commit graph, history)
- Index (Staging Area)
- Branches
- Head

git Commands

- **git add**
puts files from working directory into staging area (index). If not tracked so far, adds them to tracked files.
- **git commit**
commits files from staging area (index) to repository, moves current branch with HEAD
- **git restore [--source=<commit>] <file>**
Set <file> in working directory to state at <commit> (default: index).
- **git switch [-c] <branch>**
moves HEAD to <branch> (-c creates it), changes content of working dir
- **git reset --soft <commit>**
moves HEAD to <commit> (takes the current branch with it)
- **git reset --mixed <commit>**
moves HEAD to <commit>, changes index to be at <commit> (but not working directory)
- **git reset --hard <commit>**
moves HEAD to <commit>, changes index and working tree to <commit>.

git gui



gitk

The screenshot shows the gitk graphical user interface. At the top, a menu bar contains 'File', 'Edit', 'View', and 'Help'. The main window is divided into three panes. The left pane displays a commit graph with a vertical line of commits labeled A, B, C, F, G, and D. A yellow dot at the top represents the current commit, which is a merge of branch 'another_one' (indicated by a green box) and the master branch (indicated by a green box). The middle pane shows the commit message: 'Merge branch 'another_one' E', followed by a list of commit hashes and the committer's name, 'Andreas Mueller <t3kci@gmail.com>'. The right pane shows a list of commit hashes and their corresponding timestamps. Below the main panes, a status bar displays the SHA1 ID: 'c3ea8c81bc9b8dcf66e99c303f8e3859081124f8'. Below the status bar, a search bar is present. The bottom pane is divided into two sections: 'Diff' and 'Tree'. The 'Diff' section shows the commit message and the commit's parent and child hashes. The 'Tree' section shows the commit's parent and child hashes. The 'Diff' section also displays the commit's parent and child hashes, and the commit's parent and child hashes. The 'Tree' section shows the commit's parent and child hashes, and the commit's parent and child hashes.

File Edit View Help

master Merge branch 'another_one'
another_one E
D
G
F
C
B
A

Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>
Andreas Mueller <t3kci@gmail.com>

2017-01-23 15:43:51
2017-01-23 15:33:12
2017-01-23 15:33:05
2017-01-23 15:33:25
2017-01-23 15:33:18
2017-01-23 15:33:01
2017-01-23 15:32:56
2017-01-23 15:32:52

SHA1 ID: c3ea8c81bc9b8dcf66e99c303f8e3859081124f8 Row 5 / 8

Find commit containing: Exact All fields

Search

◆ Diff ◆ Old version ◆ New version Lines of context: 3 Ignore space change

Author: Andreas Mueller <t3kci@gmail.com> 2017-01-23 15:33:18
Committer: Andreas Mueller <t3kci@gmail.com> 2017-01-23 15:41:24
Parent: 513ced10203dfe81ecf3448b49435eacb136e561 (C)
Child: 43563a5e64b3e8daa0929e7e211ae5651ec7171b (G)
Branch: master
Follows:
Precedes:

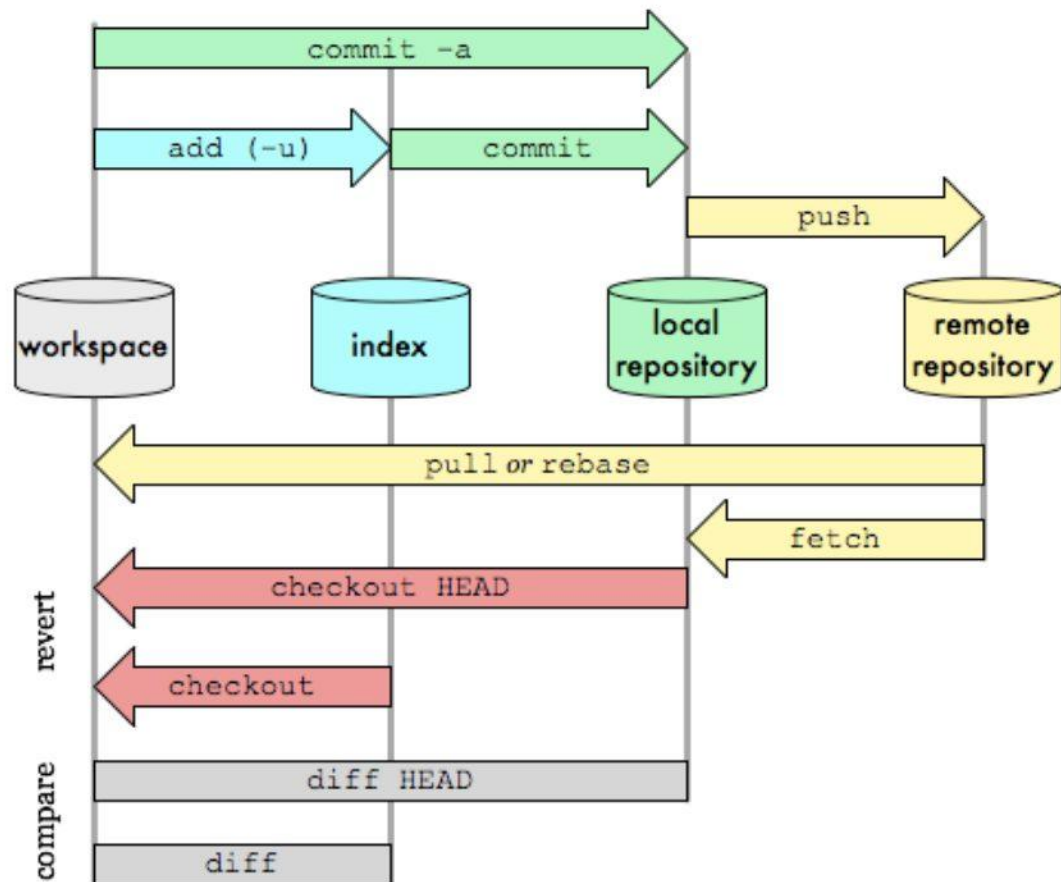
F

new file mode 100644
index 0000000..e69de29

◆ Patch ◆ Tree
Comments
F

Git Data Transport Commands

<http://osteele.com>



reflog

\$ git reflog

```
d6fedb0 HEAD@{0}: merge another_one: Merge made by the 'recursive' strategy.
43563a5 HEAD@{1}: rebase finished: returning to refs/heads/master
43563a5 HEAD@{2}: rebase: G
c3ea8c8 HEAD@{3}: rebase: F
513ced1 HEAD@{4}: rebase: checkout 513ced1
4db426c HEAD@{5}: merge feature: Fast-forward
b5ba00a HEAD@{6}: checkout: moving from master to master
b5ba00a HEAD@{7}: reset: moving to HEAD~3
6cec4ed HEAD@{8}: merge another_one: Fast-forward
513ced1 HEAD@{9}: checkout: moving from another_one to master
6cec4ed HEAD@{10}: reset: moving to 6cec4ed
4db426c HEAD@{11}: checkout: moving from feature to another_one
4db426c HEAD@{12}: checkout: moving from master to feature
513ced1 HEAD@{13}: reset: moving to HEAD~4
4db426c HEAD@{14}: checkout: moving from feature to master
4db426c HEAD@{15}: checkout: moving from master to feature
4db426c HEAD@{16}: commit: G
125e957 HEAD@{17}: commit: F
6cec4ed HEAD@{18}: commit: E
1e96a3b HEAD@{19}: commit: D
513ced1 HEAD@{20}: commit: C
b5ba00a HEAD@{21}: commit: B
db928a0 HEAD@{22}: commit (initial): A
```

```
> git log --oneline --graph --decorate --all
* d6fedb0 (HEAD -> master) Merge branch 'another_one'
| \
| * 6cec4ed (another_one) E
| * 1e96a3b D
| * 43563a5 G
| * c3ea8c8 F
|/
* 513ced1 C
* b5ba00a B
* db928a0 A
```

Git for ages 4 and up:

<https://www.youtube.com/watch?v=1ffBJ4sVUb4>

(with play-doh!)

Test-driven development?

Types of tests

- Unit tests – function does the right thing.
- Integration tests – system / process does the right thing.
- Non-regression tests – bug got removed (and will not be reintroduced).

Example

content of inc.py

```
def inc(x):  
    return x + 1
```

content of test_sample.py

```
from inc import inc
```

```
def test_answer():  
    assert inc(3) == 4
```

```
> py.test test_sample.py
```

```
===== test session starts =====
```

```
platform linux -- Python 3.5.2, pytest-2.9.2, py-1.4.31, pluggy-0.3.1
```

```
rootdir: /tmp, inifile:
```

```
plugins: cov-2.4.0, timeout-1.2.0
```

```
collected 1 items
```

```
test_sample.py .
```

```
===== 1 passed in 0.00 seconds =====
```

Test coverage

```
# inc.py
def inc(x):
    if x < 0:
        return 0
    return x + 1

def dec(x):
    return x - 1
```

```
# test_inc.py
from inc import inc

def test_inc():
    assert inc(3) == 4
```

```
> py.test
===== test session starts =====
platform linux -- Python 3.5.2, pytest-2.9.2, py-1.4.31, pluggy-0.3.1
rootdir: /tmp/myproj, inifile:
plugins: cov-2.4.0, timeout-1.2.0
collected 1 items

test_inc.py .

===== 1 passed in 0.01 seconds =====
```

Test coverage

```
# inc.py
def inc(x):
    if x < 0:
        return 0
    return x + 1

def dec(x):
    return x - 1
```

```
# test_inc.py
from inc import inc

def test_inc():
    assert inc(3) == 4
```

```
/tmp/myproj [andy@dsi-amueller] [10:51]
> py.test --cov inc
===== test session starts =====
platform linux -- Python 3.5.2, pytest-2.9.2, py-1.4.31, pluggy-0.3.1
rootdir: /tmp/myproj, inifile:
plugins: cov-2.4.0, timeout-1.2.0
collected 1 items

test_inc.py .

----- coverage: platform linux, python 3.5.2-final-0 -----
Name      Stmts  Miss  Cover
-----
inc.py      6      2    67%

===== 1 passed in 0.01 seconds =====
```

HTML report

```
$ uv run pytest --cov=inc --cov-report=html
```

Coverage report: 67%

Module ↓	statements	missing	excluded	coverage
inc.py	6	2	0	67%
Total	6	2	0	67%

coverage.py v4.2, created at 2017-01-23 10:53

Coverage for **inc.py** : 67%

6 statements 4 run 2 missing 0 excluded

```
1 # inc.py
2 def inc(x):
3     if x < 0:
4         return 0
5     return x + 1
6
7
8 def dec(x):
9     return x - 1
```

Benefits of CI

- Can run on many systems
- Can't forget to run it
- Contributor doesn't need to know details
- Can enforce style
- Can provide immediate feedback
- Protects the main branch (if run on PR)

What does it do?

- Triggered at each commit / push
- Sets up a virtual machine with your configuration.
- Pulls the current branch.
- Runs command. Usually: install, then test.
- Reports success / Failure to GitHub.

Using CI

- Triggered any time you push a change
- Integrated with pull requests
- Try a pull request on your own repository!